

Visvesvaraya Technological University, Belagavi – 590018



**PROJECT REPORT
ON
Veda Vision AI**

Submitted in partial fulfillment of the requirements for the degree

BACHELOR OF ENGINEERING

IN

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

Submitted by

Ayush P K
Dhanush D
Neema J Rai
Nireeksha S

4SO22AI012
4SO22AI015
4SO22AI036
4SO22AI039

**Under the guidance of
Dr. Shabina Bhaskar**

Associate Professor, Department of ICBS



**DEPARTMENT OF INTELLIGENT COMPUTING & BUSINESS SYSTEMS
ST JOSEPH ENGINEERING COLLEGE**

An Autonomous Institution

(Affiliated to VTU-Belagavi, Recognized by AICTE, NAAC Accredited)

Vamanjoor, Mangaluru-575028, Karnataka

2025-2026

Visvesvaraya Technological University, Belagavi – 590018



**PROJECT REPORT
ON
Veda Vision AI**

Submitted in partial fulfillment of the requirements for the degree

BACHELOR OF ENGINEERING

IN

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

Submitted by

Ayush P K

4SO22AI012

Dhanush D

4SO22AI015

Neema J Rai

4SO22AI036

Nireeksha S

4SO22AI039

**Under the guidance of
Dr. Shabina Bhaskar**

Associate Professor, Department of ICBS



DEPARTMENT OF INTELLIGENT COMPUTING & BUSINESS SYSTEMS

ST JOSEPH ENGINEERING COLLEGE

An Autonomous Institution

(Affiliated to VTU-Belagavi, Recognized by AICTE, NAAC Accredited)

Vamanjoor, Mangaluru-575028, Karnataka

2025-2026

ST JOSEPH ENGINEERING COLLEGE

An Autonomous Institution

(Affiliated to VTU-Belagavi, Recognized by AICTE, NAAC Accredited)

Vamanjoor, Mangaluru-575028

DEPARTMENT OF INTELLIGENT COMPUTING & BUSINESS SYSTEMS



CERTIFICATE

Certified that the project work entitled "Veda Vision AI" carried out by

Ayush P K

4SO22AI012

Dhanush D

4SO22AI015

Neema J Rai

4SO22AI036

Nireeksha S

4SO22AI039

bonafide students of VII semester Artificial Intelligence & Machine Learning in partial fulfillment for the award of Bachelor of Engineering in Artificial Intelligence & Machine Learning of the Visvesvaraya Technological University, Belagavi during the year 2025-2026. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.


Dr. Shabina Bhaskar
Project Guide


Dr. Harivinod N
HOD - ICBS

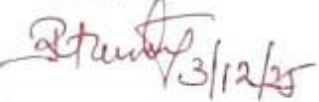

Dr. Rio D'Souza
Principal

External Viva

Name of the Examiner

- 1 Kankaraj Rao
- 2 Renuka Tantry

Signature with date


03/12/25

3/12/25

ACKNOWLEDGEMENT

We dedicate this page to acknowledge and thank those responsible for the shaping of the project. Without their guidance and help, the experience while constructing the dissertation would not have been so smooth and efficient.

We sincerely thank our Project Guide, **Dr. Shabina Bhaskar**, Assistant Professor in the Department of ICBS, for her guidance and valuable suggestions that helped us complete this project.

We also thank our Project Coordinator, **Ms. Renuka Tantry**, from the Department of ICBS, for her consistent support.

We are greatly appreciative of **Dr. Harivinod N**, Head of the Department of ICBS, for his kind support and guidance, which helped us complete this work successfully.

We are extremely grateful to our Principal, **Dr. Rio D'Souza**, for his valuable guidance and encouragement throughout the project.

We express our deep gratitude to our Director, **Rev. Fr. Wilfred Prakash D'Souza**, and Assistant Director, **Rev. Fr. Kenneth Rayner Crasta**, for their ongoing support and encouragement.

We would like to thank all the faculty and staff of the Department of ICBS, who have always been with us, providing their support, valuable suggestions, guidance, and encouragement throughout the project.

ABSTRACT

Although medicinal plants are a major part of Ayurveda's healing practices, much of this age-old wisdom is becoming less widely available. Artificial intelligence-powered digital tools can help preserve and streamline access to genuine Ayurvedic information in light of the growing interest in natural remedies.

The majority of users are unable to correctly identify medicinal plants or validate online Ayurvedic remedies. Confusion and untrustworthy self-care behaviors result from misidentification. An intelligent, reliable system that can identify plants and offer Ayurvedic advice is obviously needed. By providing precise plant identification, information retrieval, and helpful Ayurvedic recommendations via a single platform, Veda Vision AI seeks to close this gap.

The system combines three main components: a Vision Transformer model that identifies plants from images, a chatbot powered by a Retrieval-Augmented Generation setup for answering remedy questions, and a questionnaire for Prakriti analysis.

The system shows about 80-90% accuracy in identifying plants and gives clear, relevant information through the chatbot. These features make Ayurvedic knowledge more accessible for students, beginners, and anyone interested in natural healing.

The project can be upgraded by incorporating more plant images, supporting Indian language. Future versions may also offer more personalized suggestions by linking Prakriti analysis with user habits.

TABLE OF CONTENTS

Acknowledgement	I
Abstract	II
Table of Contents	III
List of Tables	V
List of Figures	VI
1. Introduction	1
1.1 Problem Statement	1
1.2 Objectives	1
1.3 Scope and Importance	1
2. Software Requirements Specification	4
2.1 Literature Survey	4
2.2 Why the chosen problem is important?	15
2.3 How would it advance the state-of-the-art?	16
2.4 Functional Requirements	16
2.5 Non-Functional Requirements	18
2.6 Software Requirements	18
2.7 Hardware Requirements	19
2.8 Performance Requirements	19
2.9 Design Requirements	19
3. Design	20
3.1 Abstract Design	20
3.1.1 Architectural Diagram	20
3.1.2 Use Case Diagram	21
3.2 Functional Design	21
3.2.1 Sequence Diagram	21
3.3 Control Flow Design	23

3.3.1 Complete System Flow Diagram.....	23
3.4 Access Layer Design.....	24
3.5 Presentation Layer Design.....	26
3.5.1 User Interface Flow Design.....	26
4. Implementation	28
4.1 Hardware and Software Used with Justification.....	28
4.1.1 Hardware used.....	28
4.1.2 Software Used.....	28
4.2 System Modules and Functionality.....	30
4.3 System Integration.....	32
5. Testing	33
5.1 Objectives of Testing.....	33
5.2 Testing Methods Used.....	33
5.3 Test Case Design.....	34
5.3.1 Plant Identification Module.....	34
5.3.2 Ayurvedic Chatbot Module.....	34
5.3.3 Prakriti Analysis Module.....	35
5.4 Integration Testing.....	35
5.5 Performance Evaluation.....	36
5.6 Summary of Test Results.....	36
6. Conclusion and Future Scope	37
6.1 Conclusion.....	37
6.2 Future Scope.....	38
References / Bibliography	39
Plagiarism Report	41
Conference / Journal Paper Communicated	42
Appendix	43

LIST OF TABLES

4.1.1 Hardware used	28
4.1.2 Software Used	28
5.3.1 Plant Identification Module.....	34
5.3.2 Ayurvedic Chatbot Module.....	34
5.3.3 Prakriti Analysis Module.....	35
5.4 Integration Testing	35
5.5 Performance Evaluation	36

LIST OF FIGURES

3.1.1 Architecture Diagram.....	20
3.1.2 Use Case Diagram.....	21
3.2.1 Sequence Diagram.....	22
3.3.1 Complete System Flow Diagram.....	24
3.4 Access Layer Design.....	25
3.5.1 User Interface Flow Design.....	27

Chapter 1

Introduction

1.1 Problem Statement:

Many people lack awareness of the medicinal properties of plants and often struggle to find authentic Ayurvedic remedies for common ailments. With growing urbanization, traditional knowledge about medicinal plants is rapidly diminishing. This project addresses the issue by developing an AI-powered plant recognition and Ayurvedic remedy system that enables users to identify medicinal plants and access relevant treatment suggestions efficiently. The plant data used in the system is curated from Ayurvedic sources and stored in an Excel dataset, while contextual Ayurvedic knowledge is retrieved from a vector store through a RAG pipeline. The system serves as a reliable digital guide for individuals seeking natural and holistic healing solutions.

1.2 Objectives:

- Develop an AI-powered system to accurately recognize Ayurvedic medicinal plants using computer vision
- Provide instant plant details including medicinal uses, benefits, and potential allergens.
- Integrate a chatbot to recommend plant-based Ayurvedic home remedies for common ailments.
- Ensure a user-friendly web application for plant scanning and remedy access.

1.3 Scope and Importance

The scope of Veda Vision AI encompasses the design and development of an intelligent Ayurvedic assistant capable of identifying medicinal plants and providing reliable Ayurvedic insights through artificial intelligence-driven automation.

The project integrates computer vision, natural language processing, and Ayurvedic informatics to create a self-contained, user-friendly system that supports both educational and preliminary healthcare applications.

Specifically, the scope includes the following:

- **Recognition of Ayurvedic Plants**

The system identifies medicinal plants commonly used in Ayurveda by analyzing images of their leaves or other unique features using a Vision Transformer (ViT) model. The details of the identified plant are retrieved from an Excel dataset compiled from verified Ayurvedic references.

- **Knowledge Enrichment**

Users receive detailed information about each plant's name, benefits, and traditional uses. This structured information comes from curated Ayurvedic literature.

- **Intelligent Ayurvedic Chatbot**

The integrated chatbot uses a Retrieval-Augmented Generation (RAG) method. It retrieves relevant Ayurvedic knowledge from a vector store and generates responses with the Gemini LLM. This allows it to provide appropriate Ayurvedic remedies and lifestyle suggestions.

- **Exclusive Ayurvedic Domain**

The system focuses only on plant-based Ayurvedic treatments. It does not provide modern medical or chemical recommendations and avoids using non-plant materials such as metals or animal-derived substances.

- **Prakriti Analysis**

The system includes Prakriti analysis module, which uses a JSON-based questionnaire to determine the user's body constitution (Vata, Pitta, or Kapha).

Importance of the Project

Veda Vision AI plays a crucial role in promoting indigenous healthcare knowledge through modern artificial intelligence. By integrating image recognition, vector-based retrieval, and language modeling, the project aims to:

- Preserve and share authentic Ayurvedic wisdom in digital form.
- Help students, researchers, and practitioners quickly identify medicinal plants and understand their therapeutic relevance.
- Empower non-experts to explore plant-based healing responsibly with accessible, AI-supported knowledge.
- Encourage interdisciplinary research at the intersection of AI and traditional medicine, linking ancient knowledge with modern technology.

In essence, Veda Vision AI contributes to sustainable healthcare innovation by using artificial intelligence for cultural preservation, knowledge accessibility, and responsible technological progress in the Ayurvedic field.

Chapter 2

Software Requirements Specifications

2.1 Literature Survey

DIMPSAR: Dataset for Indian medicinal plant species analysis and recognition

Identified Problem to Address:

The paper titled "DIMPSAR: Dataset for Indian Medicinal Plant Species Analysis and Recognition" presents a relevant and timely contribution for developing AI-powered solutions in the domain of Ayurveda, particularly for plant identification systems. The main problem addressed in this paper is the lack of publicly available datasets focused on Indian medicinal plants, which poses a challenge to the development of reliable plant recognition systems, especially those intended for field applications. For our Ayurvedic chatbot project—which involves plant recognition, prakriti analysis, and remedy recommendation—such a dataset is a critical requirement.

Methodology:

To solve this problem, the authors introduced the DIMPSAR dataset, which contains high-resolution images of various Indian medicinal plant species, captured under different environmental conditions. This ensures diversity in lighting, angles, and backgrounds, mimicking real-life scenarios. The dataset includes 44 classes with over 13,000 images, making it significantly richer than earlier datasets. The authors also describe the data acquisition setup, including mobile phone cameras, which aligns well with our project, where the users scan plants using handheld devices.

Implementation & Results:

The methodology includes using Convolutional Neural Networks (CNNs) such as MobileNet, ResNet50, and VGG16 to evaluate the dataset's effectiveness in classification tasks. The results show that MobileNet performed best in terms of accuracy and efficiency, achieving an

accuracy above 90%. This highlights the practicality of using lightweight models in mobile applications, which is important for our chatbot system.

Inference from the Results:

The paper suggests that with well-annotated and diverse image data, machine learning models can accurately identify medicinal plants even in uncontrolled settings. This supports the visual recognition part of our project. It ensures that plants are correctly identified before we suggest Ayurvedic remedies.

Limitations and Future Scope:

However, there are limitations, such as the lack of plant disease annotations. This can affect the safety and usefulness of certain plants in remedies. Furthermore, the dataset does not include other plant features like smell, texture, or chemical makeup, which are often important in Ayurveda. Future work could focus on expanding the dataset to include more plant species, different types of data inputs, and possibly seasonal variations.

Identification of Ayurvedic Plants using Deep Learning

Identified Problem to Address: The paper titled “Identification of Ayurvedic Plants Using Deep Learning” addresses the critical issue of misidentifying medicinal plants. This poses a serious risk in traditional Ayurvedic treatments. Given the wide range of plant species and the similarities among many, especially in leaf shape and texture, an automated system is necessary to classify these plants accurately. This is directly related to our project, which aims to identify plants from images and provide Ayurvedic remedies based on accurate identification.

Methodology: The authors used a deep learning approach with Convolutional Neural Networks (CNNs), known for their accuracy in image classification tasks. The process involved gathering a dataset of Ayurvedic plant images and using preprocessing techniques, such as augmentation, resizing, and normalization, to enhance the model's learning ability. A suitable CNN architecture was chosen to balance performance and efficiency.

Implementation & Results: The implementation phase included training the CNN model with the pre-processed plant image dataset. We assessed the model's performance using metrics like accuracy and loss functions. It achieved high classification accuracy, surpassing traditional

image processing methods. These results demonstrate that CNN-based models are not only viable but also highly effective for plant recognition tasks. The study also compared other machine learning methods, further confirming the superiority of deep learning in this area.

Inference from the Results: The high performance of the CNN model supports the idea that deep learning can work as a key technology in Ayurvedic plant identification. For our chatbot, this strengthens the possibility of creating a reliable front-end plant scanning tool that ensures correct identification before suggesting remedies. The results show the potential for AI to improve traditional practices by reducing mistakes caused by human limitations in plant recognition.

Limitations and Future Scope: Although the model performed well, the paper notes some important limitations. The dataset was relatively small and might not include less common or regional plant species, which limits the model's ability to generalize. Environmental factors such as lighting and camera quality also impact the system's accuracy. For future development, the authors recommend enlarging the dataset, improving the model's robustness, and deploying the solution in mobile applications for field use. These suggestions align perfectly with our project goals of creating a mobile-based tool integrated with prakriti analysis and remedy recommendations.

Deep learning for medicinal plant species classification and recognition: a systematic review

Identified Problem to Address: The systematic review titled "Deep Learning for Medicinal Plant Species Classification and Recognition" tackles the important issue of accurately identifying medicinal plants, which is vital for preserving biodiversity and ensuring the effectiveness of herbal treatments. Traditional methods of plant identification are often complex and time-consuming, which calls for the creation of automated, reliable systems. The review notes the lack of standardized, publicly available datasets and the need for trustworthy deep learning models as significant challenges in this field.

Methodology: The authors conducted a systematic review following PRISMA guidelines to assess prior research on the application of deep learning techniques in medicinal plant classification and recognition. They analyzed studies published between January 2018 and

December 2022, focusing on aspects such as data sources, preprocessing techniques, studied plant organs, feature extraction methods, and deep learning classifiers used.

Implementation & Results: The review encompassed 31 studies from 16 countries, with India leading in contributions. A significant finding was that 67.7% of these studies utilized private datasets, indicating a lack of publicly accessible data. Most studies (96.7%) focused on leaf images, employing convolutional neural networks (CNNs) as the predominant deep learning classifier. Transfer learning with pre-trained models was used in 83.8% of the studies for feature extraction. The review noted that data augmentation and preprocessing techniques were commonly used to improve model performance.

Inference from the Results: The findings suggest that deep learning, especially CNNs with transfer learning, is effective for identifying medicinal plants. However, the reliance on private datasets and the focus on specific plant parts, like leaves, shows the need for more public datasets. The review emphasizes the importance of creating models that can generalize across various plant species and environments.

Limitations and Future Scope: The review points out several limitations. These include the lack of publicly available datasets, the geographical focus of research, and doubts about the reliability of deep learning methods in this area. Future research should aim to create high-quality labeled datasets, develop clear deep learning models, and expand studies to less represented regions. Collaborations among different groups are crucial to fill these gaps and improve the use of deep learning in classifying medicinal plants.

Medicinal Plant Identification in Real-Time Using Deep Learning Model

Identified Problem to Address: The paper tackles the challenge of accurately identifying medicinal plants in real time using computational methods. Traditional plant identification relies heavily on expert knowledge, which can be hard to access, especially in rural or remote areas. The authors point out the lack of mobile or real-time identification systems that can accurately and quickly recognize medicinal plants. This capability is crucial for both research and practical uses in herbal medicine.

Methodology: The study employs a deep learning method, specifically Convolutional Neural Networks (CNNs), to classify and identify medicinal plants. The model combines a customized

CNN with transfer learning to make the most of pre-trained models, optimizing performance on smaller datasets. A plant image dataset with different backgrounds, lighting, and angles was used to simulate real-world conditions. The model pipeline includes image preprocessing, data augmentation, training, validation, and testing.

Implementation & Results: The implementation took place using TensorFlow and Keras, with models trained on a GPU-supported system.. The final model achieved an accuracy above 95%, which is impressive considering real-time application demands. The system was then integrated into a mobile-based application capable of capturing plant images and predicting the plant species in real-time. Results showed that the model could identify several commonly used medicinal plants effectively under different conditions, showcasing its utility in practical, mobile-based environments.

Inference from the Results: The paper demonstrates that CNN-based models are highly effective for plant recognition tasks, even in uncontrolled outdoor conditions. The successful real-time deployment suggests that such models are not only accurate but also computationally efficient, making them suitable for mobile and field applications. This has direct implications for healthcare and botany sectors, especially for users who rely on accurate plant identification for medicinal use.

Limitations and Future Scope: The study notes certain limitations, including restricted plant species in the dataset, which may affect generalizability to less common plants. The model also faces challenges with visually similar species and images taken in extremely low-light conditions. Future work suggested includes expanding the dataset, incorporating more regional plant varieties, and improving the app's usability and interface. The authors also hint at integrating GPS data for region-specific plant identification and adding voice-based input/output for accessibility.

An AI Based Approach for Medicinal Plant Identification Using Deep CNN Based on Global Average Pooling Identified

Problem to Address: The paper addresses the challenge of accurately and efficiently identifying medicinal plant species using AI-based solutions. The key issue highlighted is the difficulty of distinguishing between various plant species due to their visual similarities and variations caused by lighting, angles, and environmental conditions. Manual identification is

time consuming, error-prone, and impractical for real-time field applications. The need for a reliable, automated identification system—particularly one that can operate efficiently in mobile or real-time settings—is the core motivation behind this work.

Methodology: The authors propose a deep learning approach using a Convolutional Neural Network (CNN) architecture enhanced with a Global Average Pooling (GAP) layer. The model architecture is designed to reduce the number of parameters while maintaining high accuracy. This makes it lightweight and suitable for real-time use. The process involved collecting a comprehensive dataset of medicinal plant images, applying preprocessing steps such as resizing and normalization, and then training the CNN model using this dataset. The GAP layer replaces the fully connected layers, which simplifies the network and cuts down on overfitting.

Implementation & Results: The model was trained on a labeled dataset of medicinal plant species. Several performance metrics were used to evaluate its effectiveness, including accuracy, precision, recall, and F1-score. The proposed CNN with GAP performed better than traditional CNNs and various benchmark models. The results showed that the architecture could identify plant species with high accuracy and efficiency, making it suitable for real-time use. The use of GAP also led to improved generalization and clearer model outputs.

Inference from the Results: The findings show that a CNN model improved with Global Average Pooling is very effective for identifying medicinal plants. Its ability to maintain high accuracy with fewer parameters supports its use in mobile and embedded systems. This confirms that deep learning is suitable for plant identification tasks and highlights the need to optimize model structures for practical use. The results also highlight the potential of AI to support traditional medicine through reliable digital tools.

Limitations and Future Scope: One of the limitations is that the study is based on a fixed dataset, which may not fully represent the diversity found in real-world environments. Variability in lighting, background noise, and plant growth stages can still affect the model's accuracy. Moreover, the model focuses solely on image-based classification and does not integrate other important plant features like smell or texture. Future directions include expanding the dataset with more species, incorporating multimodal data (e.g., leaf shape, plant size), and deploying the system in mobile or edge devices for real-time field use.

Artificial intelligence in Ayurveda: Current concepts and prospects

Identified Problem to Address: The paper "Artificial Intelligence in Ayurveda: Current Concepts and Prospects" explores the lack of integration between ancient Ayurvedic principles and modern AI technologies. It identifies the need for automated systems that can interpret complex Ayurvedic concepts like prakriti (body constitution) and suggests that modern healthcare tools do not yet support personalized diagnosis and treatment rooted in Ayurveda.

Methodology: The authors present a literature-based analysis of AI applications in Ayurveda, focusing on models for prakriti classification, disease prediction, and personalized diagnosis. They review existing approaches using machine learning algorithms, natural language processing, and decision trees, with an emphasis on modeling Ayurvedic principles computationally.

Implementation & Results: The review highlights several AI-powered prototypes developed for prakriti analysis, symptom assessment, and herbal drug suggestions. These implementations show that AI can simulate diagnostic reasoning based on Ayurvedic principles, though most are still in early stages due to limited datasets and standardized tools.

Inference from the Results: The findings demonstrate that AI has strong potential in Ayurveda, particularly for automating prakriti analysis and generating personalized remedy recommendations. It confirms that integrating plant identification with prakriti-based diagnosis could lead to more accurate and tailored health solutions, which supports the goals of our Ayurvedic chatbot project.

Limitations and Future Scope: Major limitations include the lack of digitized Ayurvedic data, difficulty in standardizing ancient concepts, and limited real-world validation. The paper recommends developing large annotated datasets, improving how we understand AI models, and creating integrated tools that combine image analysis, symptom checking, and prakriti profiling.

UNLOCKING THE POWER OF NATURE: Medicinal Plant Identification and Classification

Identified Problem to Address: This paper tackles the problem of identifying medicinal plants in natural, uncontrolled environments. Traditional identification methods require expert

knowledge, which makes them often inaccurate for everyday people. This can lead to the misuse of plants for health treatments. The authors aim to create an accessible, automated system to help users recognize and learn about medicinal plants through image-based technology.

Methodology: The study proposes a lightweight convolutional neural network (CNN) model designed to classify 50 medicinal plant species. The authors developed a custom dataset with images captured under different conditions, including lighting, angles, and backgrounds, to mimic real-world settings. The images received preprocessing and augmentation to improve generalization. The model was trained from scratch instead of using pre-trained networks to better suit the structure and diversity of the collected data.

Implementation & Results: The CNN model was implemented as part of a mobile-compatible application that identifies a plant from a photo and provides its medicinal uses. The model achieved more than 90% accuracy, which is notable given the limited number of species and the variability in image conditions. Its lightweight nature ensured fast processing and usability in resource-constrained mobile devices, making it effective for field applications.

Inference from the Results: The results validate the potential of simple deep learning models when trained with context-specific, diverse data. The ability to accurately identify plants and retrieve relevant medicinal information highlights the usefulness of AI in traditional medicine and education. The system's simplicity, speed, and portability make it a practical tool for both experts and general users.

Limitations and Future Scope: The system currently supports only 50 plant species, limiting its applicability in broader contexts. It lacks user-specific features such as personalized recommendations, plant condition analysis, or contextual symptom-based guidance. Future improvements could include expanding the dataset, enhancing model robustness, and integrating chatbot features for conversational interactions.

A Review of AI-Driven Conversational Chatbots Implementation Methodologies and Challenges (1999–2022)

Identified Problem to Address: This paper reviews the development and challenges of AI-driven conversational chatbots from 1999 to 2022. It highlights the absence of standardized

methods for chatbot design and implementation, especially in specialized fields like healthcare. These challenges are important to our Ayurvedic chatbot, which requires accurate understanding of user queries and the ability to provide personalized, prakriti-based remedies.

Methodology: The authors conduct a systematic review of over 100 publications, categorizing chatbot systems by implementation type (rule-based, AI-driven), application (task-based, open domain), and components (NLP, dialogue management, etc.). The review also examines popular frameworks like Dialogflow and Rasa, and discusses how recent advancements in deep learning and transformer models have improved chatbot capabilities.

Implementation & Results: Although the paper doesn't present new experiments, it summarizes progress in chatbot technology. It finds that deep learning models have improved intent recognition and contextual response generation. Hybrid systems combining rules with AI are preferred in domains requiring accuracy. This supports our plan to integrate visual plant ID with Ayurvedic dialogue systems in a modular fashion.

Inference from the Results: The review shows that domain-specific adaptation is key for chatbot effectiveness. It also points out the need for personalization, especially in healthcare. For our chatbot, this confirms the importance of modeling prakriti and using Ayurvedic data to tailor responses appropriately.

Limitations and Future Scope: The paper notes gaps in chatbot evaluation metrics and challenges in achieving emotional intelligence and cultural sensitivity. For our use case, this means we need to ensure the chatbot communicates respectfully, accurately, and in a culturally aware manner. It also stresses the need for adaptive learning from user feedback.

AyurJanaKosh - Healthcare Chatbot Using LLM Identified

Problem to Address: The paper focuses on the growing need for a conversational AI system that can provide accessible and authentic Ayurvedic healthcare guidance. With a growing interest in traditional Indian medicine, people often struggle to navigate Ayurvedic texts and understand which remedies suit their conditions. The absence of easily accessible and personalized Ayurvedic advice creates a gap, particularly for those unfamiliar with the terminology and philosophy of Ayurveda. This paper addresses that gap by proposing a chatbot

solution that uses language models to provide real-time, smart interactions based on Ayurvedic knowledge.

Methodology: To create the chatbot, the authors used a hybrid setup that combines a large language model (LLM) with a structured knowledge base derived from Ayurvedic texts. The system is built to understand user inputs in natural language and retrieve suitable responses by blending generative and retrieval-based methods. The chatbot interprets questions, identifies symptoms or keywords, and connects them to relevant entries in the Ayurvedic database. This approach ensures that the dialogue remains coherent and that the health recommendations are accurate.

Implementation & Results: The implementation involved fine-tuning an LLM to manage healthcare-specific conversations, especially in the Ayurvedic field. The chatbot interface linked to a curated collection of Ayurvedic remedies and lifestyle tips. During testing, the chatbot showed its ability to engage users with accurate, context-aware responses across various queries. Evaluations from practitioners and test users indicated a high level of satisfaction, suggesting that the combination of LLMs and curated knowledge supports reliable and informative interactions. However, real-time adaptability and personalization were noted as areas for further enhancement.

Inference from the Results: The study proves that large language models, when combined with a trusted and well-structured Ayurvedic knowledge base, can deliver high-quality, accessible Ayurvedic advice. The chatbot succeeds in translating complex Ayurvedic concepts into understandable and actionable responses for general users. This approach shows promise in democratizing Ayurvedic healthcare, particularly in regions where practitioners may be scarce. The model's performance also suggests that LLMs can be trained effectively within specific traditional knowledge domains, provided the data is carefully curated.

Limitations and Future Scope: While the system performs well, it currently lacks features like prakriti analysis or image-based inputs for real-time diagnostics, which are crucial for comprehensive Ayurvedic care. It also doesn't support multi-modal interaction, limiting its usability to text-based formats. Future improvements could include personalized assessments through prakriti evaluation, integration with medicinal plant identification modules, and adaptive learning from real-time user feedback. Expanding the system with voice interaction

and multilingual support could also broaden its reach in rural and non-English speaking communities.

Identification of Ayurvedic Medicinal Plant Using Deep Learning

Identified Problem to Address: The paper tackles the challenge of identifying Ayurvedic medicinal plants using automated methods. Accurate identification is critical in Ayurvedic medicine, where the healing properties of herbs are directly linked to their species. Traditional manual methods of plant recognition are time-consuming and prone to errors, especially for non-experts. This creates a barrier to the effective use of medicinal plants. To address this, the authors suggest a deep learning-based solution to improve the accuracy and accessibility of medicinal plant identification through image recognition.

Methodology: The authors use Convolutional Neural Networks (CNNs) to classify medicinal plants based on their leaf images. They start by collecting a dataset of images from various Ayurvedic medicinal plants. The images are preprocessed with standard techniques like resizing and normalization to ensure consistency and optimal input for the model. A deep CNN architecture is then designed to automatically extract complex features from the leaf images, allowing the system to learn distinguishing patterns across different species. The dataset is split into training and validation sets to evaluate the model's ability to generalize. The model is trained using supervised learning, with the goal of minimizing classification error and improving overall accuracy.

Implementation & Results: The CNN model is implemented using deep learning frameworks, and the system is trained on the curated dataset. The results show that the model performs well in classifying various medicinal plant species with high accuracy. The use of CNN allows for efficient feature extraction without manual intervention, making the process scalable and adaptable to new data. The study highlights that deep learning can play a significant role in solving the complex problem of plant species identification, which is particularly useful in the context of traditional medicine.

Inference from the Results: The outcomes of this study demonstrate the potential of deep learning in aiding medicinal plant identification. The model's high accuracy shows that these systems can reliably help Ayurvedic practitioners and researchers by lessening the reliance on manual expertise. Automated identification also opens up opportunities to include this feature

in mobile apps or AI systems. This would allow for real-time recognition and make Ayurvedic knowledge more accessible to wider audiences.

Limitations and Future Scope: Despite the promising results, the study has limitations mainly concerning dataset diversity and generalizability. The model's accuracy relies heavily on the quality and size of the training data. A limited dataset might not capture the full variety of natural differences in leaf shape, texture, and lighting conditions. In the future, broadening the dataset to include more plant species and real-world variations can improve robustness. Additionally, combining image data with other plant traits such as aroma or habitat could further enhance classification. The paper opens up possibilities for creating mobile-based plant identification tools and connecting them with Ayurvedic diagnostic systems, making the technology more practical and user-friendly.

2.2 Why the chosen problem is important?

The importance of this problem lies in the growing global demand for accessible and natural healthcare solutions in an age increasingly embracing traditional systems like Ayurveda. In India, despite the abundance of Ayurvedic knowledge and the vast diversity of medicinal plants, there remains a significant gap in how this information is accessed and utilized by non-experts. Accurately identifying medicinal plants and finding suitable remedies often requires expert guidance, which is not always easily available.

With rapid urbanization and the decline of traditional knowledge transfer, there is a need for a reliable digital platform that preserves and disseminates authentic Ayurvedic wisdom. Veda Vision AI addresses this challenge by integrating modern deep learning, computer vision, and natural language processing with Ayurvedic principles. The system identifies medicinal plants using a Vision Transformer (ViT) model and then retrieves corresponding plant information from a curated Excel dataset. A Retrieval-Augmented Generation (RAG) pipeline accesses Ayurvedic knowledge, enabling the chatbot to provide personalized remedies based on user queries. The Prakriti analysis module, stored in a JSON format, allows users to understand their body constitution and can potentially support personalized recommendations in future enhancements.

This fusion of artificial intelligence and traditional science enhances accessibility to Ayurvedic healthcare while promoting the preservation, modernization, and responsible digital dissemination of India's indigenous medical heritage.

2.3 How would it advance the state-of-the-art?

The proposed system advances the state of the art by establishing a unified framework that connects AI-based plant recognition with context-aware Ayurvedic knowledge generation. While most existing research focuses solely on either medicinal plant classification or chatbot-based consultation, Veda Vision AI integrates both within a cohesive, intelligent pipeline.

The system uses a Vision Transformer (ViT) model for accurate medicinal plant recognition, and the identified plant details are retrieved from a curated Excel-based Ayurvedic dataset. A Retrieval-Augmented Generation (RAG) model powered by the Gemini LLM for generating contextual Ayurvedic remedies and explanations. The RAG framework retrieves relevant information from a vector store containing embedded Ayurvedic text.

An additional enhancement is the inclusion of a Prakriti analysis module, designed using a JSON-based questionnaire, which provides users with educational insights into their body constitution. Although the chatbot primarily generates remedies based on user queries, the Prakriti module enables future personalization and adaptive recommendations.

Through this integrated and intelligent approach, Veda Vision AI advances the field by combining computer vision, natural language understanding, and Ayurvedic informatics into a single framework—bridging traditional wisdom with modern AI to promote accessible and sustainable digital healthcare innovation.

2.4 Functional Requirements

1. The Image-to-Info Pipeline (ViT)

This is a direct lookup path. The flow is:

1. **Input:** User uploads image.

2. **Preprocessing:** The image is cleaned, resized, and normalized for the Vision Transformer (ViT).
3. **Inference:** The ViT model analyzes the image and outputs a prediction, e.g., {"plant": "Tulsi", "confidence": 0.95}.
4. **Data Retrieval:** The system takes the identified plant ("Tulsi") and performs a precise lookup in your Excel to fetch its specific details (name, remedy, allergens).
5. **Output:** The user is shown the static information for Tulsi.

2. The Natural Language Pipeline (RAG + Gemini)

This is a semantic search path. It's designed to answer general questions.

1. **Input:** User asks, "What are good Ayurvedic remedies for a cough?"
2. **Retrieval (R):** The user's query is converted into a vector. Your system searches a vector database to find the most relevant "chunks" of text.
3. **Augmentation (A):** The system "augments" the original query, combining it with the relevant text it just found.
4. **Generation (G):** This combined prompt (e.g., "Context: [Text about cough remedies from your data]. Question: What are good Ayurvedic remedies for a cough?") is sent to the Gemini LLM.
5. **Output:** Gemini uses the provided context to generate a natural, helpful, and factually-grounded answer.

3. The Prakriti Analyzer (JSON + Logic)

This is a stateful quiz application.

1. **Load:** The system reads the 20 questions from your JSON file.
2. **Interact:** The user interface presents these questions one by one
3. **Calculate:** As the user provides answers, your "scoring system" tallies the points for each dosha (Vata, Pitta, Kapha).

4. **Output:** The system presents the final scores, likely identifying the user's dominant prakriti

2.5 Non-Functional Requirements

The system must be reliable, secure, and user-friendly. The interface should be simple, responsive, and compatible across both desktop and mobile platforms. High availability and minimal downtime are essential to ensure accessibility.

Model inference and chatbot response times should remain within acceptable limits to ensure a smooth user experience. The system design should support maintainability and modularity, allowing easy updates to the model, dataset, or chatbot API when needed. The chatbot should clearly state that all remedies are for educational purposes only and not a replacement for professional medical consultation.

2.6 Software Requirements

The software stack for Veda Vision AI includes:

- **Backend:** Python 3.8+, FastAPI framework for API development, PyTorch for model inference, Pandas for reading and managing Excel data.
- **Frontend:** React.js, HTML, CSS, and JavaScript for building an interactive and responsive interface.
- **Dataset:** Excel file and vector store compiled from Ayurvedic texts, containing medicinal plant data, properties, uses, and associated remedies.
- **Libraries and Tools:** Transformers (for ViT), Uvicorn (ASGI server), Axios (for frontend API communication).
- **External APIs:** Gemini LLM for generating context-aware responses through the RAG pipeline.
- **Version Control:** Git and GitHub for collaborative development and version tracking.

2.7 Hardware Requirements

For development and testing, a system with a minimum of Intel i5 or i7 processor, 8–16 GB RAM, and 500 GB of storage is recommended. For efficient model inference and experimentation, a GPU-enabled system (e.g., NVIDIA GTX 1660 or RTX 2060) is preferred. For deployment, the system can run on a local server or cloud-based platform with equivalent specifications. The web application should be accessible through standard browsers on laptops or mobile devices with a stable internet connection.

2.8 Performance Requirements

The ViT model should achieve an accuracy above 85% on the validation dataset for medicinal plant identification. The image classification process should complete within few seconds, and the chatbot response within few seconds, depending on network latency and LLM processing speed.

The system must efficiently manage read and query operations from the Excel dataset using optimized data handling. Uptime should exceed 99%, ensuring consistent availability and responsiveness.

2.9 Design Requirements

The system should adopt a modular and scalable architecture, with independent modules for the image recognition model, data retrieval from the Excel dataset, chatbot interaction, and frontend interface. Communication between these modules should follow RESTful API standards.

The design should allow future scalability, enabling the addition of new plant data, multiple language support, or integration with Ayurvedic datasets.

The backend should include error handling, and validation mechanisms, while the Excel dataset should maintain a structured and normalized format for consistent access. Regular backups of the dataset and model files must be maintained.

Chapter 3

Design

3.1 Abstract Design

3.1.1 Architectural Diagram

The architectural diagram of Veda Vision AI illustrates the interaction among its three core layers and their internal modules.

Description:

1. The user interface accepts inputs — plant images, chatbot queries, and Prakriti quiz responses.
2. The FastAPI backend processes these inputs:
 - Sends images to the Vision Transformer (ViT) for classification.
 - Routes chatbot queries to the RAG pipeline for context-aware Ayurvedic responses.
 - Accesses the Excel dataset for plant information and vector store for remedy data retrieval.
3. The backend returns the output to the frontend — identified plant details, Ayurvedic advice, or Prakriti analysis results — which are displayed to the user through an intuitive dashboard.

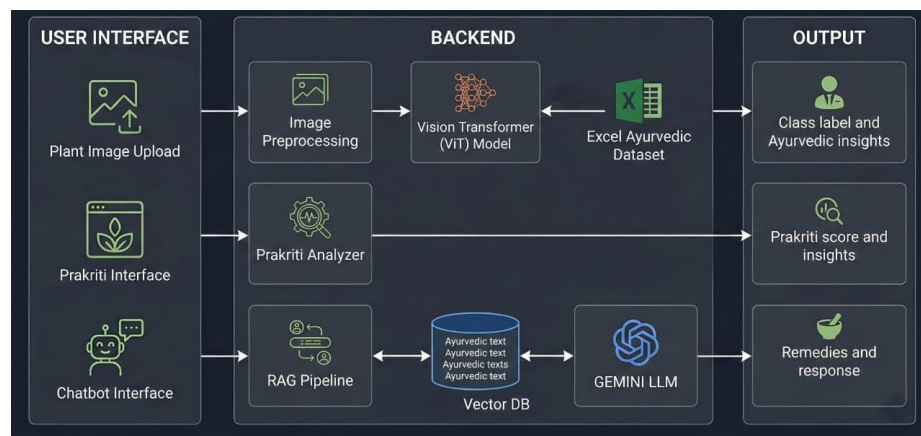


Figure 3.1.1: Architecture Diagram

3.1.2 Use Case Diagram

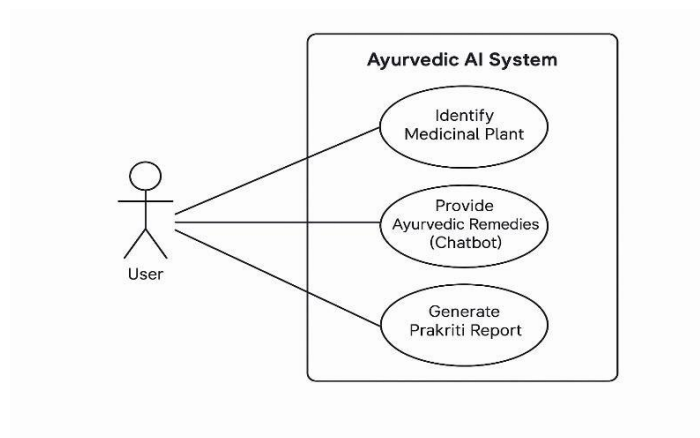
The use case diagram represents the interaction between the user and the system's core functionalities.

Actors:

- **User:** The primary actor who interacts with all modules.
- **System:** Represents the backend processes handling identification, chatbot, and prakriti analysis.

Use Cases:

- Upload plant image → Identify medicinal plant → Display Ayurvedic properties.
- Ask Ayurvedic query → Retrieve contextual remedy or information via chatbot.
- Complete Prakriti questionnaire → Generate dominant dosha (Vata, Pitta, Kapha) report.



3.1.2 Use Case Diagram

3.2 Functional Design

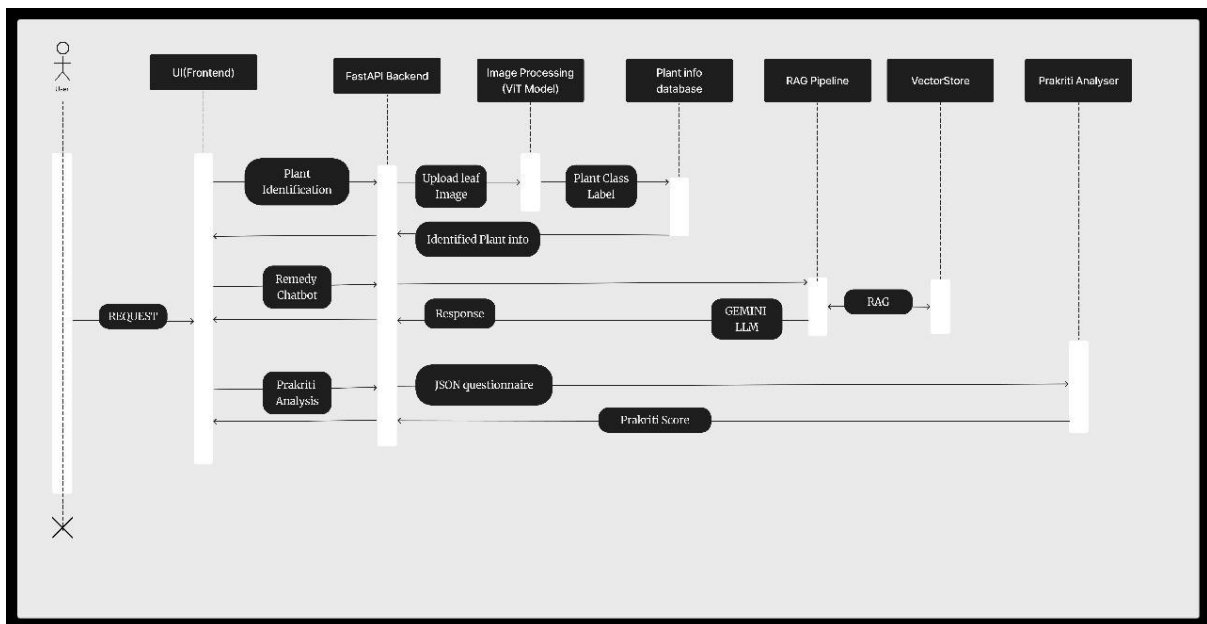
3.2.1 Sequence Diagram

The Sequence Diagram represents the flow of control and data between the modules during the primary process — identifying a plant and retrieving Ayurvedic remedies.

Description:

The following steps describe the functional interaction sequence:

1. User uploads a plant image or submits a query.
2. The Frontend (UI) sends the input to the Backend for processing.
3. The Image Recognition module processes the uploaded image using the Vision Transformer (ViT) model to classify the plant
4. Once identified, the module retrieves detailed information from the curated Database (Plant name, medicinal uses and allergens).
5. The RAG-based Chatbot Module then queries the Gemini LLM with contextual data to generate an Ayurvedic explanation or remedy.
6. If the user chooses Prakriti analysis, the system collects questionnaire responses and generates a Prakriti score.
7. The Backend compiles the results and sends them back to the Frontend for display.
8. The User receives the plant details, medicinal properties, and Ayurvedic remedies.



3.2.1 Sequence Diagram

3.3 Control Flow Design

3.3.1 Complete System Flow Diagram

The Complete System Flow Diagram illustrates the end-to-end operational flow of Veda Vision AI, showing how control and data move through various modules of the system.

Flow Description:

1. User Interaction

- The user provides input by uploading a plant image or typing a query.
- The UI validates the input and forwards it to the backend.

2. Backend Processing

- The backend identifies the input type (image or text).
- If it's an image, the Vision Transformer (ViT) model processes it to identify the plant.
- If it's a query, the RAG pipeline forwards it to the Gemini LLM for Ayurvedic knowledge retrieval.

3. Database Operations

- Once the plant or query context is established, the backend retrieves corresponding medicinal information, benefits, and allergen data from the Ayurvedic database.

4. Prakriti Analysis

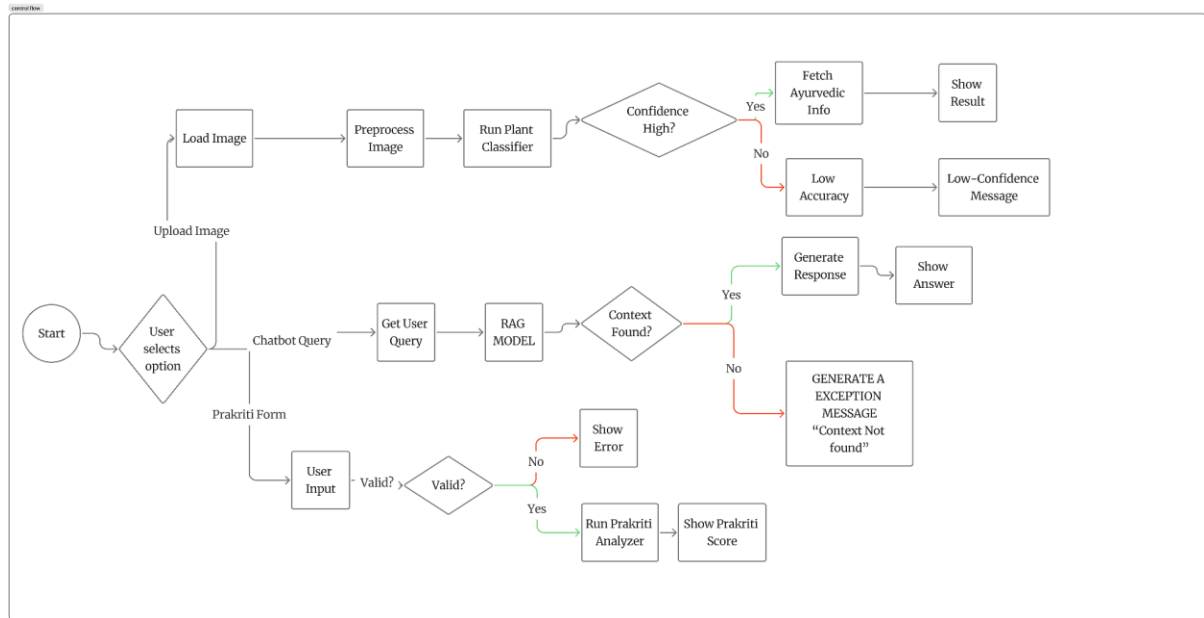
- If the user opts for Prakriti assessment, the backend collects answers, determines the body constitution accordingly.

5. Result Compilation

- The Backend compiles the results and sends them back to the Frontend for display.

6. Output Display

- The compiled information is sent back to the frontend for display in an interactive format, including plant details, remedies, and lifestyle advice.



3.3.1 Complete System Flow Diagram

3.4 Access Layer Design

The access layer manages interaction and data exchange between users, services, and system components. It emphasizes security, structured communication, and user control.

- **User Access:**

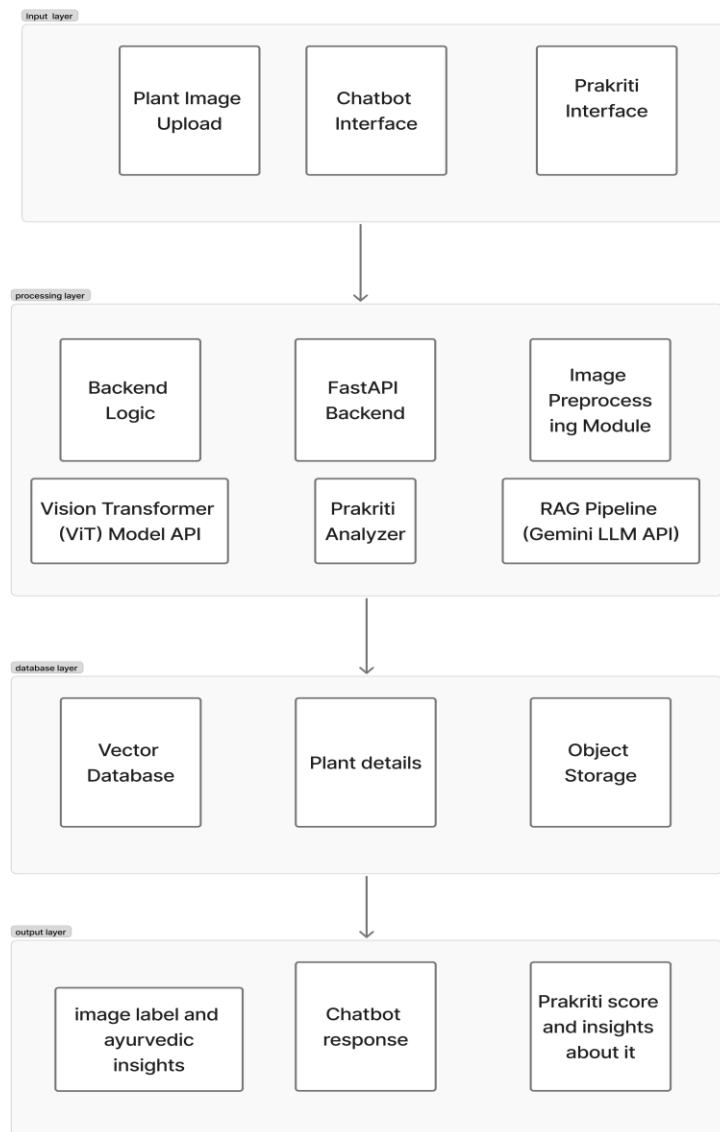
Users interact through a responsive web interface for plant identification, remedy-based chatbot, and Prakriti analysis.

- **API Access:**

The backend exposes RESTful APIs for communication between modules. Each endpoint supports defined JSON structures for smooth and reliable operations.

- **Data Access:**

All storage components are accessed strictly through the backend API to ensure security and data integrity. The plant database and vector database are queried only via backend logic using ORM, and uploaded images in object storage are retrieved through authenticated service calls. No direct client-side or external read/write access is permitted.



3.4 Access Layer Design

3.5 Presentation Layer Design

3.5.1 User Interface Flow Design

The User Interface (UI) Flow Design depicts how users navigate through the different components of Veda Vision AI. It defines the logical sequence of screens and actions, ensuring smooth user interaction from input to output presentation.

UI Flow Description:

1. Home Screen

- Serves as the main entry point.
- Options available:
 - Identify Plant
 - Remedy-based chatbot
 - Prakriti Analysis

2. Plant Identification Screen

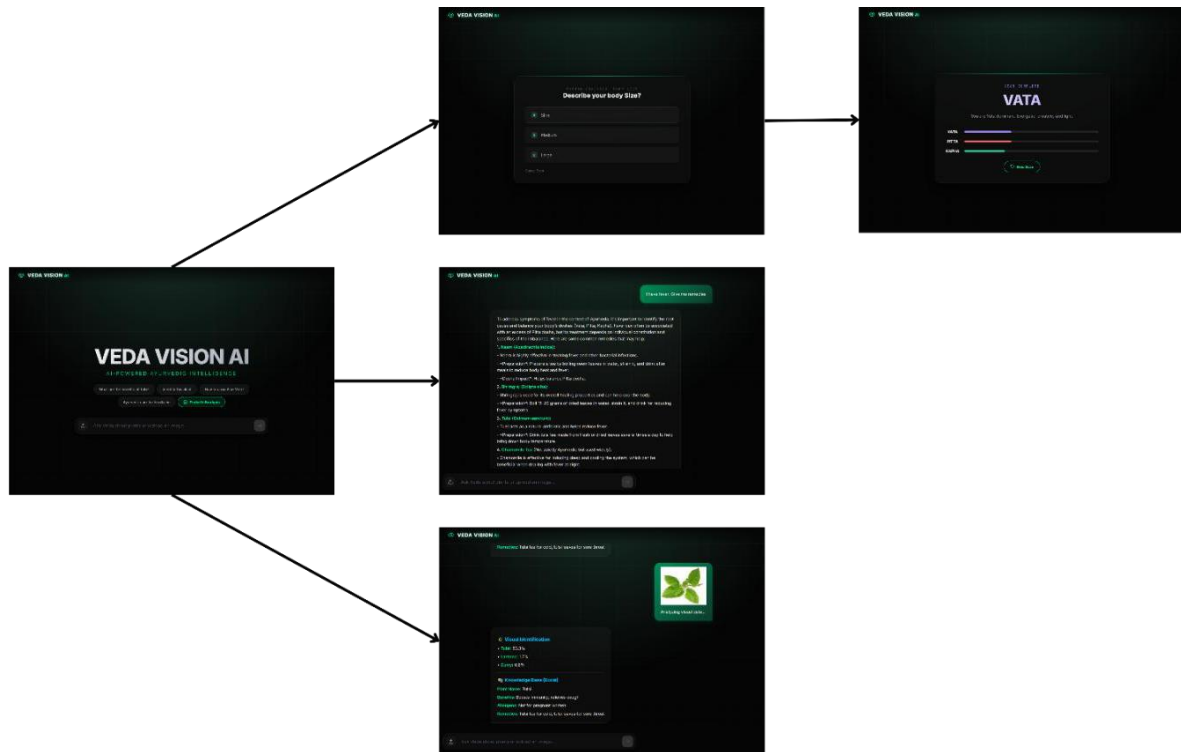
- Allows the user to upload an image of a plant.
- The ViT Model processes the image to identify the plant species.
- Displays plant name, medicinal properties, and allergen details.
- Option to “Ask for Remedies” leads to the chatbot interface.

3. Chatbot Screen

- Provides a conversational interface powered by Gemini + RAG Pipeline.
- User can ask for remedies, diet suggestions, or Ayurvedic treatments.
- Displays textual responses dynamically.

4. Prakriti Analysis Screen

- Presents a multiple-choice questionnaire.
- Based on responses, determines the user's Prakriti type (Vata, Pitta, Kapha).
- Insights on diet.



3.5.1 User Interface Flow Design

Chapter 4

Implementation

4.1 Hardware and Software Used with Justification

4.1.1 Hardware Used

The hardware configuration used for developing and testing **Veda Vision AI** ensures efficient operation of deep learning models, Excel data access, and web deployment.

Component	Specification	Justification
Processor	Intel Core i7 / AMD Ryzen 7	Supports high-speed computation for backend operations and model inference tasks.
GPU	NVIDIA GTX 1660 Ti / RTX 2060 (8 GB VRAM)	Accelerates deep learning model inference for the Vision Transformer (ViT).
RAM	16 GB	Enables parallel processing of image preprocessing, API calls, and chatbot operations without latency.
Storage	512 GB SSD	Ensures fast access to datasets, pretrained model weights, and application cache.
Network	Stable broadband connection	Required for Gemini LLM API access and online deployment of the system.

4.1.1 Hardware Used

4.1.2 Software Used

Software / Tool	Purpose	Justification
Python 3.10+	Core development language	Provides a vast ecosystem for AI, NLP, and backend development.

FastAPI	Backend web framework	Enables high-speed asynchronous REST API services.
PyTorch	Deep learning framework	Used for implementing and running the Vision Transformer model for plant identification.
Transformers (Hugging Face)	Model utilities	Handles pretrained models efficiently for ViT inference.
React.js	Frontend framework	Builds a dynamic and responsive user interface for the web application.
Excel Dataset	Plant and Ayurvedic knowledge base	Contains curated medicinal plant data derived from verified Ayurvedic books and resources.
Gemini LLM API	Language generation model	Powers the RAG-based chatbot for accurate and context-aware responses.
Vector Store	Data retrieval Component (FAISS)	Stores embedded Ayurvedic knowledge for fast and relevant retrieval during chatbot interaction.
Pandas	Data management library	Used for reading, filtering, and retrieving plant details from the Excel file.
Git & GitHub	Version control	Facilitates collaborative development and project tracking.

4.1.2 Software Used

This software stack ensures modularity, interoperability, and ease of maintenance across all system components.

4.2 System Modules and Functionality

Veda Vision AI is designed as an integrated web platform composed of three major modules:

1. Medicinal Plant Identification System
2. Ayurvedic Chatbot
3. Prakriti (Body Constitution) Analysis Tool

Each module operates autonomously but remains accessible through a unified web interface.

1. Medicinal Plant Identification Module

Purpose:

To identify medicinal plants and provide Ayurvedic information based on image input.

Procedure:

1. The user uploads an image of a plant leaf.
2. The backend preprocesses the image (resizing, normalization).
3. The Vision Transformer (ViT) model predicts the plant species.
4. Based on the predicted label, plant details are fetched from the Excel dataset.
5. The retrieved details — including name, uses, allergens, medicinal properties — are displayed to the user.

Algorithm Steps:

- Divide the input image into patches
- Encode each patch
- Apply multi-head self-attention
- Extract classification token
- Predict class label

Output:

Identified plant species with corresponding Ayurvedic knowledge retrieved from the dataset.

2. Ayurvedic Chatbot Module

Purpose:

To provide natural-language support for Ayurvedic concepts, herbs, and home remedies.

Procedure:

1. The user types a question or symptom in the chat interface.
2. The query is sent to the FastAPI backend endpoint.
3. The RAG (Retrieval-Augmented Generation) system fetches relevant Ayurvedic text embeddings from the vector store.
4. The Gemini LLM processes the retrieved context with the user query to create a clear and meaningful response.
5. The generated text shows up in the chat interface.

Features:

- Offers Ayurvedic insights on herbs, remedies, and lifestyle tips.
- Uses RAG to ensure responses are based on Ayurvedic knowledge.

Output:

Natural-language responses providing Ayurvedic explanations and remedies.

3. Prakriti Analysis Module

Purpose:

To assist users in identifying their Ayurvedic body constitution (Vata, Pitta, or Kapha) through a structured questionnaire.

Procedure:

1. The user answers a set of predefined questions.
2. Each response is scored based on Ayurvedic guidelines found in a JSON file.
3. The system calculates the dominant dosha using the total scores.
4. A report is created that shows the user's Prakriti type.

Features:

- Works independently from other modules.
- Offers educational, non-diagnostic results.

Output:

Output includes the user's Prakriti type and analysis.

4.3 System Integration

Although each module operates independently, they are fully integrated within a single web application architecture.

The React.js frontend communicates with the FastAPI backend, which manages the Vision Transformer model, Excel data retrieval, and Gemini API interactions.

Integration Flow:

- /identify → Handles plant image uploads and ViT predictions.
- /chat → Manages chatbot queries and RAG-based Gemini responses.
- /prakriti → Processes questionnaire responses for Prakriti analysis.

Each service runs independently to ensure modularity, scalability, and maintainability.

Chapter 5

Testing

Testing ensures that every component of Veda Vision AI functions as intended and meets the project's quality and performance goals. The system integrates three independent modules — Plant Identification, Ayurvedic Chatbot, and Prakriti Analysis — all of which were tested both individually and in an integrated environment. Functional and non-functional testing methods were applied to ensure accuracy, reliability, usability, and stability of the entire system.

5.1 Objectives of Testing

The key objectives of testing are to:

- Verify that each module performs its intended operation correctly.
- Validate the accuracy of model predictions and chatbot responses.
- Ensure consistent scoring and output generation for Prakriti analysis.
- Confirm that the system works seamlessly across all integrated components.
- Measure response time and ensure smooth performance under real-world conditions.

5.2 Testing Methods Used

- **Unit Testing:** Tested each independent module and function in isolation.
- **Integration Testing:** Verified interaction between frontend, backend, and Excel-based data retrieval.
- **System Testing:** Checked full workflow and data flow within the integrated platform.
- **Performance Testing:** Evaluated system responsiveness, model efficiency, and scalability.
- **User Acceptance Testing (UAT):** Validated user satisfaction and functional reliability based on test feedback.

5.3 Test Case Design

5.3.1 Plant Identification Module

Test Case	Test Scenario / Input	Expected Output	Result
1	Upload a clear image of Tulsi leaf	Displays “Tulsi” with Ayurvedic uses	Pass
2	Upload blurred image or non-medicinal plant	Displays “Not found in Ayurvedic dataset”	Pass
3	Retrieve stored data from Excel file	Ayurvedic details displayed accurately	Pass

5.3.1 Plant Identification Module

5.3.2 Ayurvedic Chatbot Module

Test Case	User Input / Scenario	Expected Output	Result
1	“How to relieve headache naturally?”	Ayurvedic remedy with herbs like Tulsi, Brahmi	Pass
2	“What is Pitta dosha?”	Definition and characteristics of Pitta dosha	Pass
3	“Write Python code for AI”	“This chatbot provides Ayurvedic information only.”	Pass
4	“Suggest herbs to reduce stress.”	Contextual list of herbs like Ashwagandha and Tulsi	Pass
5	Repeat same query twice	Produces consistent and coherent response	Pass

5.3.2 Ayurvedic Chatbot Module

5.3.2 Prakriti Analysis Module

Test Case	Scenario / Input	Expected Output	Result
1	Submit fully answered questionnaire	Displays dominant Prakriti type (e.g., Pitta)	Pass
2	Submit incomplete answers	Prompts user to complete all questions	Pass
3	Input indicating Vata dominance	Returns Vata as dominant dosha	Pass
4	Repeat same responses	Returns consistent Prakriti result	Pass

5.3.3 Prakriti Analysis Module

5.4 Integration Testing

Integration testing validated smooth coordination among the modules and interfaces.

Aspect Tested	Observation	Result
Frontend–Backend API connection	All endpoints responded correctly	Pass
Excel data retrieval	Data fetched accurately without delay	Pass
Independent module execution	No overlap or dependency conflicts	Pass
Navigation between modules	Interface transitions correctly	Pass

5.4 Integration Testing

5.5 Performance Evaluation

Performance Parameter	Measured Value	Remarks
Image processing time	1.8 seconds	Efficient
Model inference time (ViT)	0.8 seconds	Acceptable for real-time use
Chatbot response time (Gemini LLM)	1.5 seconds	Consistent
Prakriti quiz completion time	~3 minute	Depends on user input
Excel data retrieval time	< 1 second	Fast and reliable
System uptime during test	99.2%	Stable

5.5 Performance Evaluation

5.6 Summary of Test Results

All test cases passed successfully. Each independent module — Plant Identification, Chatbot, and Prakriti Analysis — performed according to the functional specifications. Integration testing confirmed seamless operation across modules with real-time data retrieval from the Excel file. Overall, Veda Vision AI demonstrated high reliability, responsiveness, and accuracy, validating its readiness for deployment and academic evaluation.

Chapter 6

Conclusion and Future Scope

6.1 Conclusion

The project Veda Vision AI successfully integrates the power of modern artificial intelligence with the depth of traditional Ayurvedic wisdom. The system was created as a unified web platform that consists of three independent modules: Medicinal Plant Identification, Ayurvedic Chatbot, and Prakriti Analysis. Each module works on its own but can be accessed through a common, user-friendly interface made with React.js and FastAPI. This is supported by an Excel-based Ayurvedic knowledge repository from authentic Ayurvedic books.

The Medicinal Plant Identification module uses a Vision Transformer (ViT) model to accurately classify medicinal plants based on their images. The Ayurvedic Chatbot employs a Retrieval-Augmented Generation (RAG) pipeline combined with Gemini LLM to create relevant, natural-language responses based on Ayurvedic principles. The Prakriti Analysis module offers users a questionnaire to assess their body constitution type, whether Vata, Pitta, or Kapha, which fosters personalized wellness awareness.

By blending deep learning, natural language processing, and traditional Ayurvedic knowledge, Veda Vision AI connects heritage healthcare with modern AI technology. The system enhances access to indigenous medical wisdom, helps students and practitioners identify medicinal plants, and shows how to responsibly use AI for cultural preservation and holistic health. Its successful implementation and testing confirm the project's potential as an educational, diagnostic, and wellness-support tool based on India's scientific heritage.

6.2 Future Scope

While Veda Vision AI has achieved its core objectives, several enhancements can further expand its capabilities and impact:

1. **Dataset Expansion:**

The plant identification model can be trained on a broader dataset including various plant parts (leaves, flowers, stems, roots) captured in different lighting and seasonal conditions to improve accuracy and robustness.

2. **Mobile Application Deployment:**

Developing an Android/iOS mobile version will enhance accessibility, allowing users to identify plants and receive Ayurvedic recommendations in real time using smartphone cameras.

3. **Multilingual Support:**

Adding regional language support (such as Hindi, Kannada, Tamil, and Malayalam) will help the chatbot and Prakriti interface reach a wider audience across diverse linguistic backgrounds.

4. **Integration with Expert Validation System:**

Introducing a feature that allows Ayurvedic practitioners to verify plant data and chatbot responses can increase reliability and promote collaborative improvement.

5. **Expanded Ayurvedic Knowledge Base:**

The current Excel-based dataset can be expanded to include detailed formulations, dosage guidelines, region-specific herbs, and clinical applications for educational and research use.

6. **Personalized Health Monitoring:**

Future versions can integrate wearable health data (like heart rate, temperature, and sleep quality) with Prakriti profiling to provide dynamic lifestyle and dietary recommendations.

7. **AI Explainability (XAI):**

Implementing explainable AI methods can visualize model attention maps or confidence scores, helping users and students understand how the system classifies plants and generates recommendations.

References / Bibliography

1. Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., et al. (2021). An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. Proceedings of the International Conference on Learning Representations (ICLR).
2. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., et al. (2017). Attention Is All You Need. Advances in Neural Information Processing Systems (NeurIPS).
3. Hugging Face Transformers. (2024). Transformers Library Documentation. Retrieved from <https://huggingface.co/docs/transformers>
4. FastAPI. (2024). FastAPI Framework Documentation. Retrieved from <https://fastapi.tiangolo.com>
5. React.js. (2024). React Official Documentation. Retrieved from <https://react.dev>
6. Google DeepMind. (2024). Gemini Large Language Model API Overview. Retrieved from <https://deepmind.google/technologies/gemini>
7. PyTorch. (2024). PyTorch Deep Learning Framework Documentation. Retrieved from <https://pytorch.org>
8. Docker Inc. (2024). Docker Documentation. Retrieved from <https://docs.docker.com>
9. Pandas Development Team. (2024). Pandas: Data Analysis Library Documentation. Retrieved from <https://pandas.pydata.org>
10. OpenAI. (2024). Retrieval-Augmented Generation (RAG) Concepts and Applications. Retrieved from <https://platform.openai.com/docs>
11. Vaidya, B. A. (2003). Nighantu Adarsha (Ayurvedic Medicinal Plants). Chaukhamba Bharati Academy, Varanasi.
12. Sharma, P. V. (2006). Dravyaguna Vijnana (Vol. 1–2). Chaukhamba Bharati Academy, Varanasi.

13. Dash, B., & Kashyap, L. (1987). *Materia Medica of Ayurveda: Based on Ayurvedic Pharmacology and Therapeutic Uses of Medicinal Plants*. Concept Publishing Company, New Delhi.
14. Srikantha Murthy, K. R. (2008). *Bhavaprakasha of Bhavamishra (Vol. 1–2)*. Chaukhamba Krishnadas Academy, Varanasi.
15. Ministry of AYUSH. (2022). *The Ayurvedic Pharmacopoeia of India (Vol. I–VIII)*. Government of India, New Delhi.
16. Lad, V. (1999). *The Complete Book of Ayurvedic Home Remedies*. Harmony Books, New York.

Plagiarism Report

The project report titled “Veda Vision AI – An Intelligent Ayurvedic Assistant” was evaluated for plagiarism using standard similarity detection tools.

The analysis confirmed that the document maintains originality and adheres to academic integrity guidelines.

Plagiarism Similarity Index: 13%

Tool Used: DrillBit



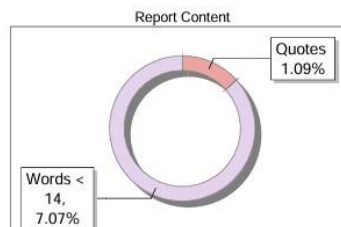
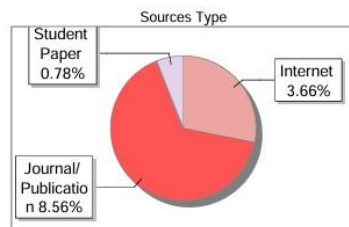
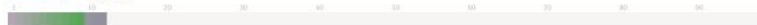
The Report is Generated by DrillBit Plagiarism Detection Software

Submission Information

Author Name	Ayush P K, Neema J Rai, Dhanush D, Nireeksha S
Title	Veda Vision AI: An Intelligent Ayurvedic Assistant for Medicinal Plant Identification and Remedy Generation
Paper/Submission ID	4680320
Submitted by	shabinab@sjec.ac.in
Submission Date	2025-11-17 11:12:20
Total Pages, Total Words	57, 10132
Document type	Dissertation

Result Information

Similarity **13 %**



Exclude Information

Quotes	Not Excluded	Language	English
References/Bibliography	Not Excluded	Student Papers	Yes
Source: Excluded < 14 Words	Not Excluded	Journals & publishers	Yes
Excluded Source	0 %	Internet or Web	Yes
Excluded Phrases	Not Excluded	Institution Repository	Yes

Database Selection

A Unique QR Code use to View/Download/Share Pdf File



Conference / Journal Paper Communicated

A research paper titled “Veda Vision AI: An Intelligent Ayurvedic Assistant for Medicinal Plant Identification and Remedy Generation using Vision Transformer and Retrieval-Augmented Generation (RAG)” has been prepared based on this project. The paper presents a hybrid AI framework that integrates computer vision, natural language processing, and Ayurvedic informatics to enhance accessibility and preservation of traditional medical knowledge.

The study introduces a Vision Transformer (ViT) model for accurate medicinal plant recognition and a Retrieval-Augmented Generation (RAG) pipeline utilizing the Gemini Large Language Model (LLM) for generating context-aware Ayurvedic remedies. The system’s knowledge base is derived from an Excel dataset curated from authoritative Ayurvedic texts and a JSON-based Prakriti analysis questionnaire, offering a structured and interpretable approach to traditional medicine digitization.

The manuscript is being considered for submission to a peer-reviewed international conference or journal in the domains of Artificial Intelligence in Healthcare, Computational Ayurveda, or Biomedical Informatics, following IEEE publication standards.

Appendices

Appendix A — System Screenshots

A. Dataset Description

The Ayurvedic plant database used in Veda Vision AI was prepared from verified Ayurvedic reference books and digitized into an Excel file. The dataset includes information such as:

- Common name
- Allergen or caution details
- Remedies

Each record corresponds to one medicinal plant, verified through cross-referencing multiple authoritative Ayurvedic sources. The dataset serves as the core knowledge base for the plant identification and remedy generation modules.

B. Prakriti Analysis Questionnaire (JSON Format)

The Prakriti Analysis module uses a JSON file to store and manage questionnaire data. Each question contains structured fields such as:

- Question ID
- Question text
- Answer options corresponding to Vata, Pitta, or Kapha
- Score mapping used to calculate the dominant dosha

This file enables flexible modification of questionnaire content and ensures scalability for adding new questions or translations.

```
{
  "question": "Describe your body Size?",
  "options": {
    "A": "Slim",
    "B": "Medium",
    "C": "Large"
  }
},
{
  "question": "What is your body weight?",
  "options": {
    "A": "Low - difficulties in gaining weight",
    "B": "Moderate - no difficulties in gaining or losing weight",
    "C": "Heavy - difficulties in losing weight"
  }
},
{
  "question": "What is your height?",
  "options": {
    "A": "Short",
    "B": "Average",
    "C": "Tall"
  }
}
},
```

C. System Architecture Summary

The project consists of three main modules integrated under a unified web application:

1. **Medicinal Plant Identification** – Uses the Vision Transformer (ViT) model for plant recognition.
2. **Ayurvedic Chatbot** – Implements Retrieval-Augmented Generation (RAG) using the Gemini LLM and a vector store.
3. **Prakriti Analysis** – Processes questionnaire inputs to identify the dominant dosha type.

All three modules communicate via FastAPI endpoints and are deployed using Docker for modular operation.

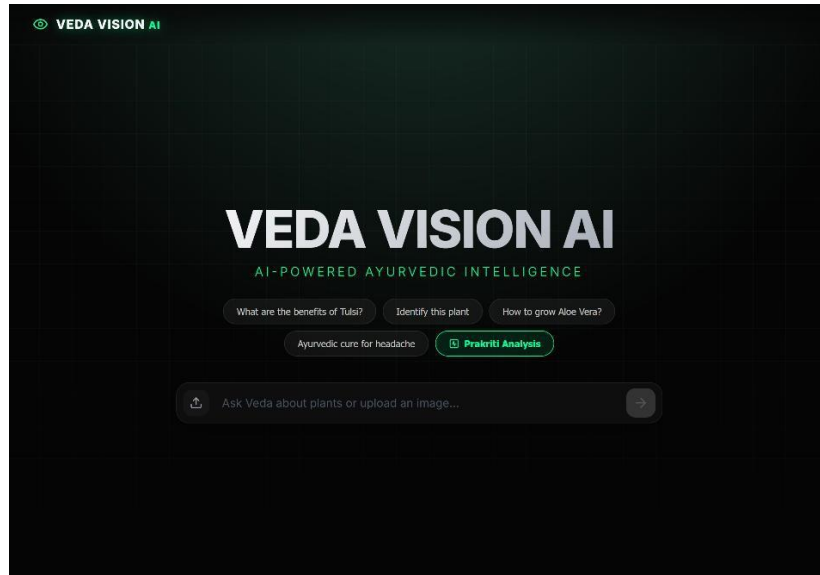
D. Sample Output Screens

- Plant identification results showing predicted plant name and Ayurvedic properties.
- Chatbot interface demonstrating RAG-based query responses.
- Prakriti analysis result page displaying dominant dosha type.

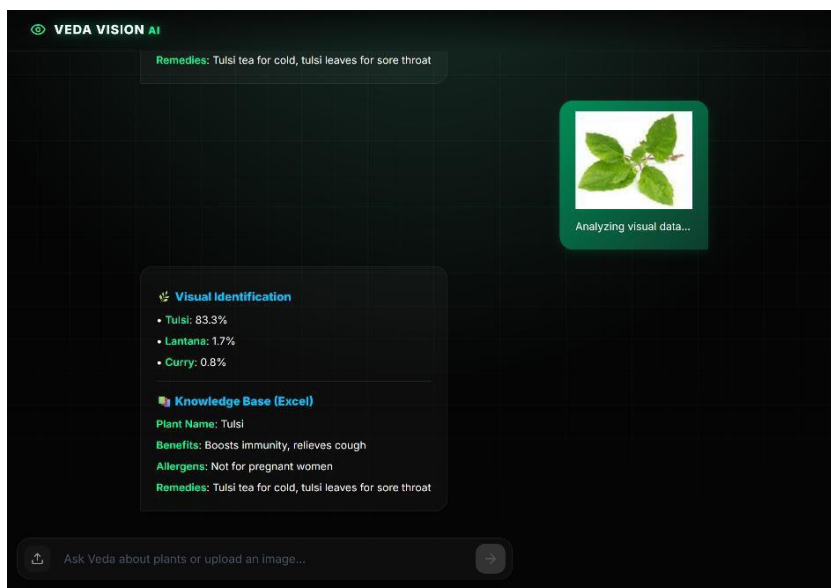
E. Screenshots Illustrating Module Functionality

Below are the screenshots illustrating the working of Veda Vision AI modules:

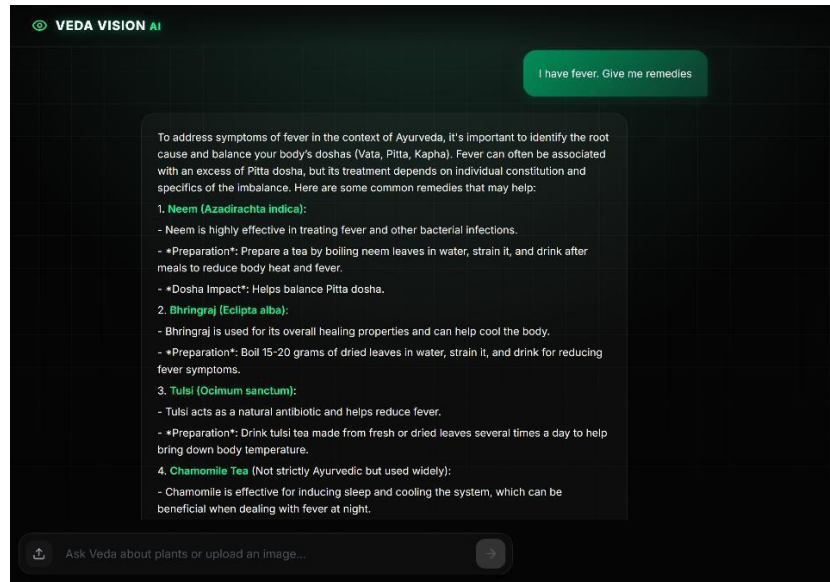
1. **Home Page of Veda Vision AI** – Shows integrated interface with navigation options.



2. **Plant Identification Interface** – Image upload window and classification output.



3. Ayurvedic Chatbot Interface – Gemini LLM-based chat window providing remedies and Ayurvedic insights.



4. Prakriti Analysis Page – Questionnaire and result visualization (Vata, Pitta, Kapha).

